CSE 270 Sports Analytics

Measuring individual performance in team sports

Habet Madoyan

July 14, 2018

Plus/Minus

Plus/Minus

The concept

Plus-minus measures the point differential for the team when the player is on the court,

Point differential is the difference in the points scored for and the points scored against the team.

The NHL's Montreal Canadiens were the first team to track the plus - minus of its players, starting sometime in the 1950s. Other teams followed in the early 1960s, and the NHL started officially compiling the statistic for the 1967 - 68 season.

Plus/minus for NBA

```
colnames(nba12)
```

```
##  [1] "GameID"           "Date"             "AwayT"            "HomeT"
##  [5] "StartTime"        "EndTime"          "ElapsedTime"      "ElapsedSecs"
##  [9] "HomePlayer1ID"    "HomePlayer2ID"    "HomePlayer3ID"    "HomePlayer4ID"
## [13] "HomePlayer5ID"    "AwayPlayer1ID"    "AwayPlayer2ID"    "AwayPlayer3ID"
## [17] "AwayPlayer4ID"    "AwayPlayer5ID"    "HomePlayer1Name"  "HomePlayer2Name"
## [21] "HomePlayer3Name"  "HomePlayer4Name"  "HomePlayer5Name"  "AwayPlayer1Name"
## [25] "AwayPlayer2Name"  "AwayPlayer3Name"  "AwayPlayer4Name"  "AwayPlayer5Name"
## [29] "StartScoreHome"   "StartScoreAway"   "EndScoreHome"     "EndScoreAway"
## [33] "PointsScoredHome" "PointsScoredAway" "PossessionsHome"  "PossessionsAway"
```

Plus/minus for NBA

Subseting the data frame for Oklahoma City Thunders

```
oks_g <- grepl("OKC", nba12$GameID, fixed = TRUE)
okc<-nba12[oks_g == T,]
```

```
head(okc[,19:23], n=5)
```

```
##     HomePlayer1Name HomePlayer2Name HomePlayer3Name HomePlayer4Name
## 144       Westbrook       Sefolosha          Durant           Ibaka
## 145       Westbrook       Sefolosha          Durant           Ibaka
## 146       Westbrook          Harden          Durant           Ibaka
## 147       Westbrook          Harden          Durant           Ibaka
## 148       Westbrook          Harden          Durant           Ibaka
##     HomePlayer5Name
## 144         Perkins
## 145         Perkins
## 146         Perkins
## 147        Collison
## 148        Collison
```

Plus/minus for NBA

Create the line-ups variable

```
okc$line_h <-
  paste(okc$HomePlayer1Name, okc$HomePlayer2Name,     okc$HomePlayer3Name, okc$HomeP
        okc$HomePlayer5Name, sep = ",")

okc$line_a <-
  paste(okc$AwayPlayer1Name, okc$AwayPlayer2Name, okc$AwayPlayer3Name,okc$AwayPlayer
        okc$AwayPlayer5Name,   sep = ",")

okc$line_h[1:10]
```

```
##  [1] "Westbrook,Sefolosha,Durant,Ibaka,Perkins"
##  [2] "Westbrook,Sefolosha,Durant,Ibaka,Perkins"
##  [3] "Westbrook,Harden,Durant,Ibaka,Perkins"
##  [4] "Westbrook,Harden,Durant,Ibaka,Collison"
##  [5] "Westbrook,Harden,Durant,Ibaka,Collison"
##  [6] "Westbrook,Harden,Durant,Ibaka,Collison"
##  [7] "Maynor,Harden,Durant,Collison,Mohammed"
##  [8] "Maynor,Harden,Durant,Collison,Mohammed"
##  [9] "Maynor,Cook,Harden,Collison,Mohammed"
## [10] "Maynor,Cook,Harden,Collison,Mohammed"
```

Plus/minus for NBA

Calculating plus/minus variable for time stunts

Away team

```
# Home team plus minus
okc$Home_PM <- okc$PointsScoredHome - okc$PointsScoredAway
# Away team plus minus
okc$Away_PM <- -okc$Home_PM
```

Check if Durant is playing

```
# is Durant playing at home?
okc$D_H <- grepl("Durant", okc$line_h, fixed = TRUE)
# is Durant playing away?
okc$D_A <- grepl("Durant", okc$line_a, fixed = TRUE)
```

Plus/minus for NBA

Get the plus minus for Durant for the whole season

```
sum(okc$Home_PM[okc$D_H == T]) + sum(okc$Away_PM[okc$D_A == T])
```

```
## [1] 324
```

This will be the plus/minus for the whole season

Plus/minus for NBA

For each game

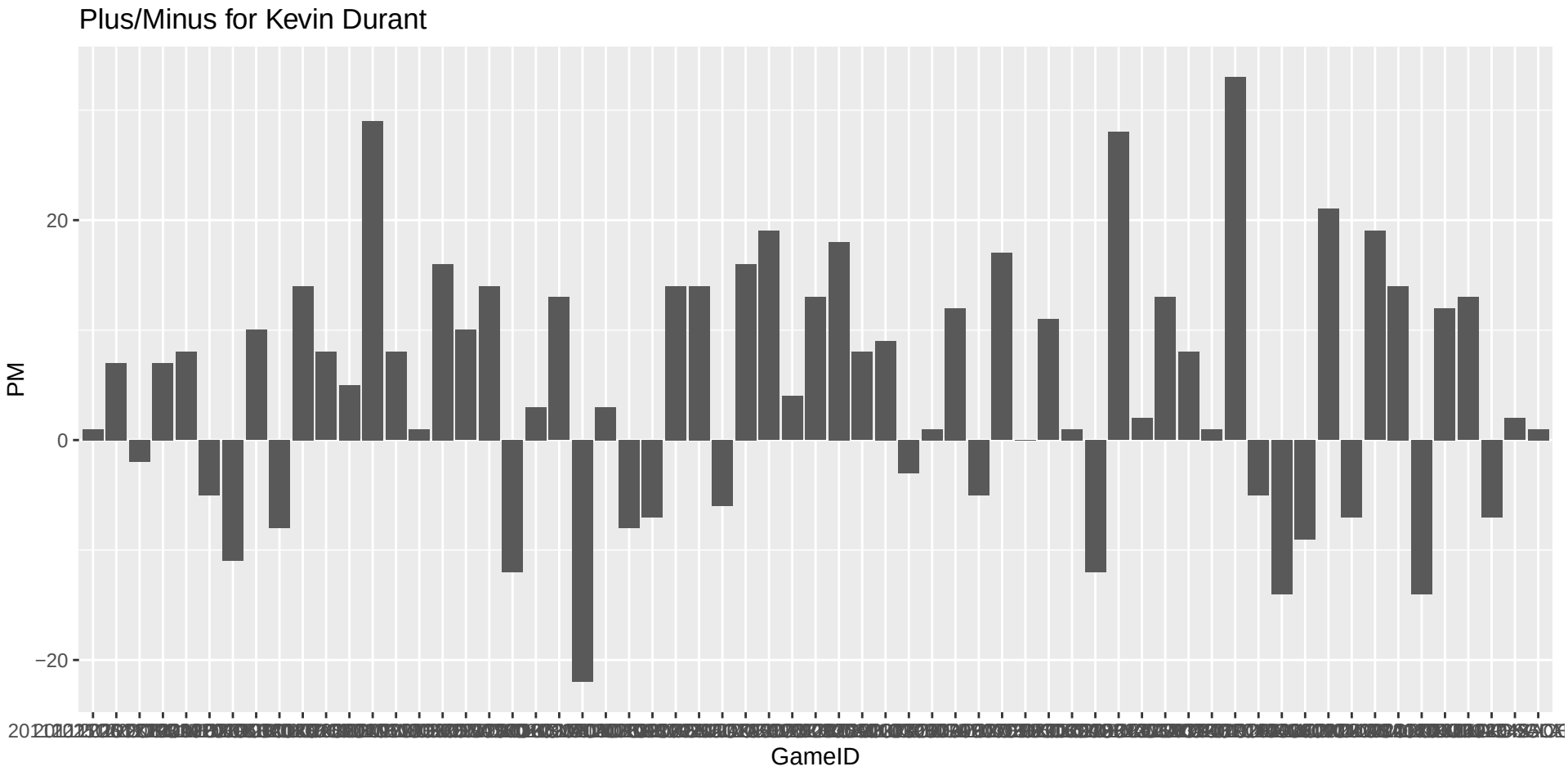
```
durant_pm <- okc %>%
  mutate(PM=ifelse(D_H==T,
                   Home_PM,
                   ifelse(D_A==T, Away_PM,0))) %>%
  group_by(GameID) %>%
  summarise(PM=sum(PM))
```

```
head(durant_pm, n=5)
```

```
## # A tibble: 5 x 2
##   GameID              PM
##   <fct>            <dbl>
## 1 20111225ORLOKC       1
## 2 20111226OKCMIN       7
## 3 20111228OKCMEM      -2
## 4 20111229DALOKC       7
## 5 20111231PHXOKC       8
```

Plus/minus for NBA

```
ggplot(durant_pm, aes(x=GameID, y=PM))+
  geom_bar(stat="identity")+
  ggtitle("Plus/Minus for Kevin Durant")
```



On court and off court +/- and Roland's rating

The on court +/- number represents the team's net points with the player on the floor per 48 minutes, while the off court number is the team's net with the player off the floor per 48 minutes.

The Roland Rating is the difference between the two, with a positive number indicating the team has played better with the player than without.

Net Plus/minus for NBA

Create a variable that shows OKC plus/minus, depending on whether it is playing at home or away

```
okc <- okc %>%
# makes PM for OKC
mutate(OKC_PM = ifelse(AwayT == "OKC", Away_PM, ifelse(HomeT == "OKC", Home_PM, 0)),
       D_PM = ifelse(D_H == T | D_A == T, T, F)
       )
```

Net Plus/minus for NBA

For each game, D_PM has a value of FALSE if Durant has not played, and TRUE if He has. PM is showing the plus/minus for that game

```
durant_pm <- okc %>%
group_by(GameID, D_PM) %>%
summarise(PM = sum(OKC_PM))
head(durant_pm)
```

```
## # A tibble: 6 x 3
## # Groups:   GameID [3]
##   GameID         D_PM     PM
##   <fct>          <lgl> <dbl>
## 1 20111225ORLOKC FALSE     7
## 2 20111225ORLOKC TRUE      1
## 3 20111226OKCMIN FALSE    -3
## 4 20111226OKCMIN TRUE      7
## 5 20111228OKCMEM FALSE     5
## 6 20111228OKCMEM TRUE     -2
```

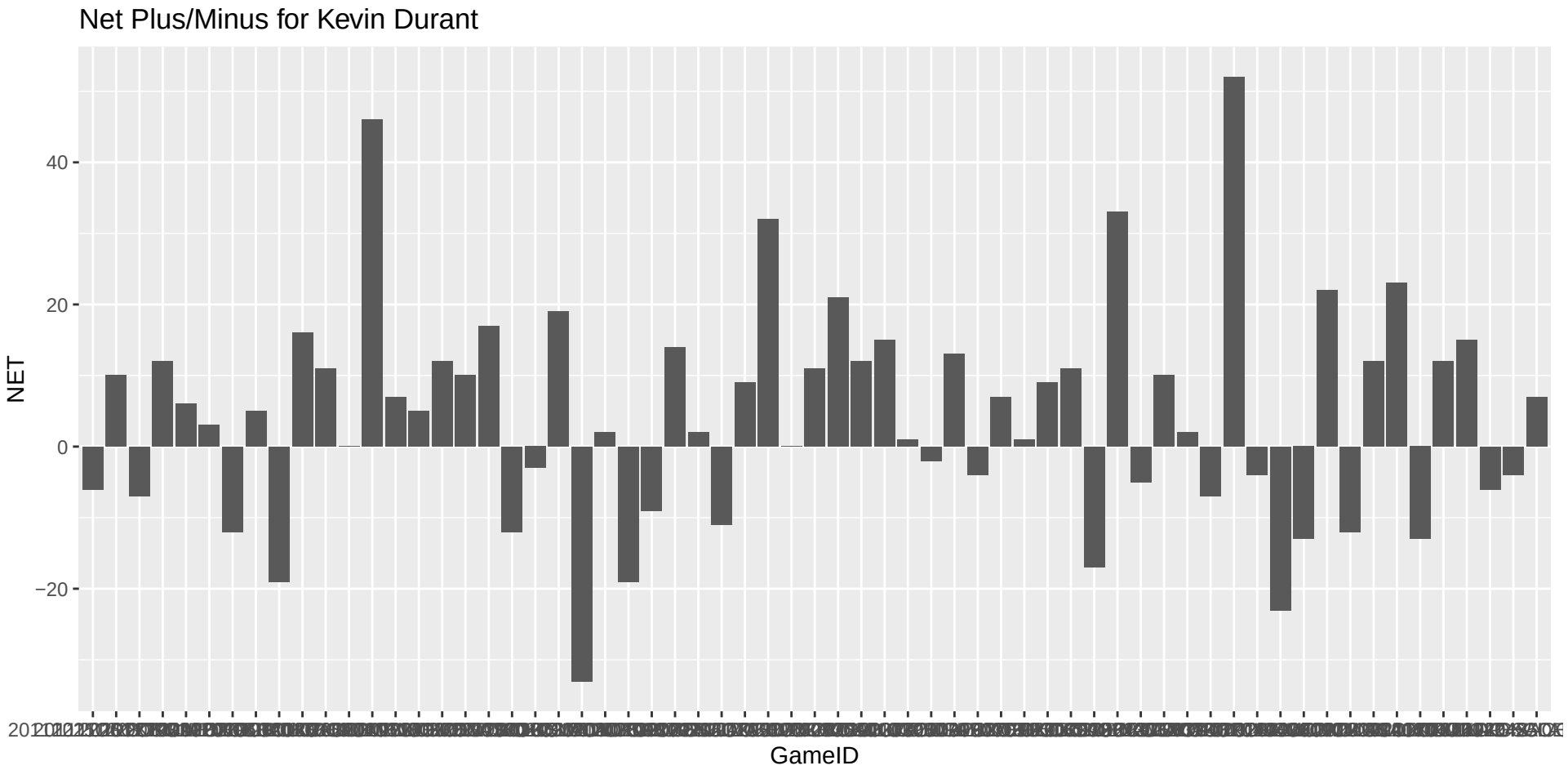
Net Plus/minus for NBA

```
durant_pm <- okc %>%
group_by(GameID, D_PM) %>%
summarise(PM = sum(OKC_PM)) %>%
group_by(GameID) %>%
mutate(NET = PM - lag(PM, 1)) %>%
filter(NET != "NA") %>%
select(GameID, NET)
head(durant_pm)
```

```
## # A tibble: 6 x 2
## # Groups:   GameID [6]
##   GameID           NET
##   <fct>          <dbl>
## 1 20111225ORLOKC    -6
## 2 20111226OKCMIN    10
## 3 20111228OKCMEM    -7
## 4 20111229DALOKC    12
## 5 20111231PHXOKC     6
## 6 20120102OKCDAL     3
```

Plus/minus for NBA

```
ggplot(durant_pm, aes(x=GameID, y=NET))+
  geom_bar(stat="identity")+
  ggtitle("Net Plus/Minus for Kevin Durant")
```



Adjusted plus/minus

Adjusted plus/minus

Adjusted plus/minus

The plus/minus rating is adjusted by the players who are on the court

$$MARGIN = \beta_0 + \beta_1 X_1 + \beta_2 X_@ + ...\beta_k X_k + \epsilon$$

where

$X_k = 1$, if player k is playing at home,
$X_k = -1$, if player k is playing away,
$X_k = 0$, if player k is not playing

Adjusted plus/minus

According to Dan Rosenbaum's approach, β coefficients will show the adjusted plus/minus of the players, or

β_k measures the difference between player k and the reference players, holding the other k players constant

Data preparation/ this will take at least 30-40 minutes on the average machine

Data prep for adjusted plus/minus

```
# Get home and away teams
nba3 <- nba12[, c("HomePlayer1ID", "HomePlayer2ID",
                  "HomePlayer3ID", "HomePlayer4ID", "HomePlayer5ID")]

nba4 <- nba12[, c("AwayPlayer1ID", "AwayPlayer2ID",
                  "AwayPlayer3ID", "AwayPlayer4ID","AwayPlayer5ID")]

# MAke ID as character
nba3 <- as.data.frame(sapply(nba3, as.character))
nba4 <- as.data.frame(sapply(nba4, as.character))

#create dummy variables
nba3 <- as.data.frame(model.matrix(~. - 1, nba3))
nba4 <- as.data.frame(model.matrix(~. - 1, nba4))
# for away team
nba4[nba4==1]<- -1
# change colnames to be more readable
colnames(nba3) <- gsub("HomePlayer[123456789]ID", "", colnames(nba3))
colnames(nba4) <- gsub("AwayPlayer[123456789]ID", "", colnames(nba4))

# nba3 remove double entries
nba3 <- t(nba3)     # transpose the data.frame
nba3 <- by(nba3, INDICES=row.names(nba3), FUN=colSums) # collapse the rows with the same name
nba3 <- as.data.frame(do.call(cbind,aggr))
nba4 <- t(nba4)     # transpose the data.frame
nba4 <- by(nba4, INDICES=row.names(nba4), FUN=colSums) # collapse the rows with the same name
nba4 <- as.data.frame(do.call(cbind,nba4))

#combine home and away players and double columns
nba5<-data.frame(nba3,nba4)
nba5 <- t(nba5)     # transpose the data.frame
nba5 <- by(nba5, INDICES=row.names(nba5), FUN=colSums) # collapse the rows with the same name
nba5 <- as.data.frame(do.call(cbind,nba5))
```

Adjusted plus/minus

```
load("./data/nba5.rda")
```

```
dim(nba5)
```

```
## [1] 25300   474
```

Plus/minus for NBA

Add plus/minus as dependent variable

```
nba5<-data.frame(PM=nba12$PointsScoredHome-nba12$PointsScoredAway, nba5)
```

Run the regression

```
model<-lm(PM~., data=nba5)
coeff<-sort(coefficients(model), decreasing=T)
coeff[1:10]
```

```
##     X1027      X453      X936      X678      X900      X158      X904      X781
## 2.1176032 1.8681832 1.5567575 1.0007997 0.9188331 0.8786333 0.8486712 0.8390129
##      X919      X964
## 0.8211618 0.8188564
```

Remove X from the begining of the coeff name and match with player info

```
name1<-names(coeff)
name1<-gsub("X", "", name1)
name1<-as.data.frame(name1)
```

Plus/minus for NBA

```
load("./data/nba12p.rda")
head(nba12p[,1:3],5)
```

```
##   PlayerID   PlayerTrueName PlayerTeams
## 1        3   Dampier, Erick         ATL
## 2       17   Diop, DeSagana         CHA
## 3       35    Boykins, Earl         HOU
## 4       36  Najera, Eduardo         CHA
## 5       42 Elson, Francisco         PHI
```

Plus/minus for NBA

```
nba12p$PlayerID<-as.character(nba12p$PlayerID)
name2<-left_join(name1, nba12p[,1:2], by=c("name1"="PlayerID"))
name2[1:5,]
```

```
##   name1    PlayerTrueName
## 1  1027     Benson, Keith
## 2   453        Diogu, Ike
## 3   936   N'Diaye, Hamady
## 4   678    Williams, Sean
## 5   900 Whiteside, Hassan
```

rename coefficients by names

```
names(coeff)<-name2$PlayerTrueName
```

Lets look at top ten players

```
coeff[1:10]
```

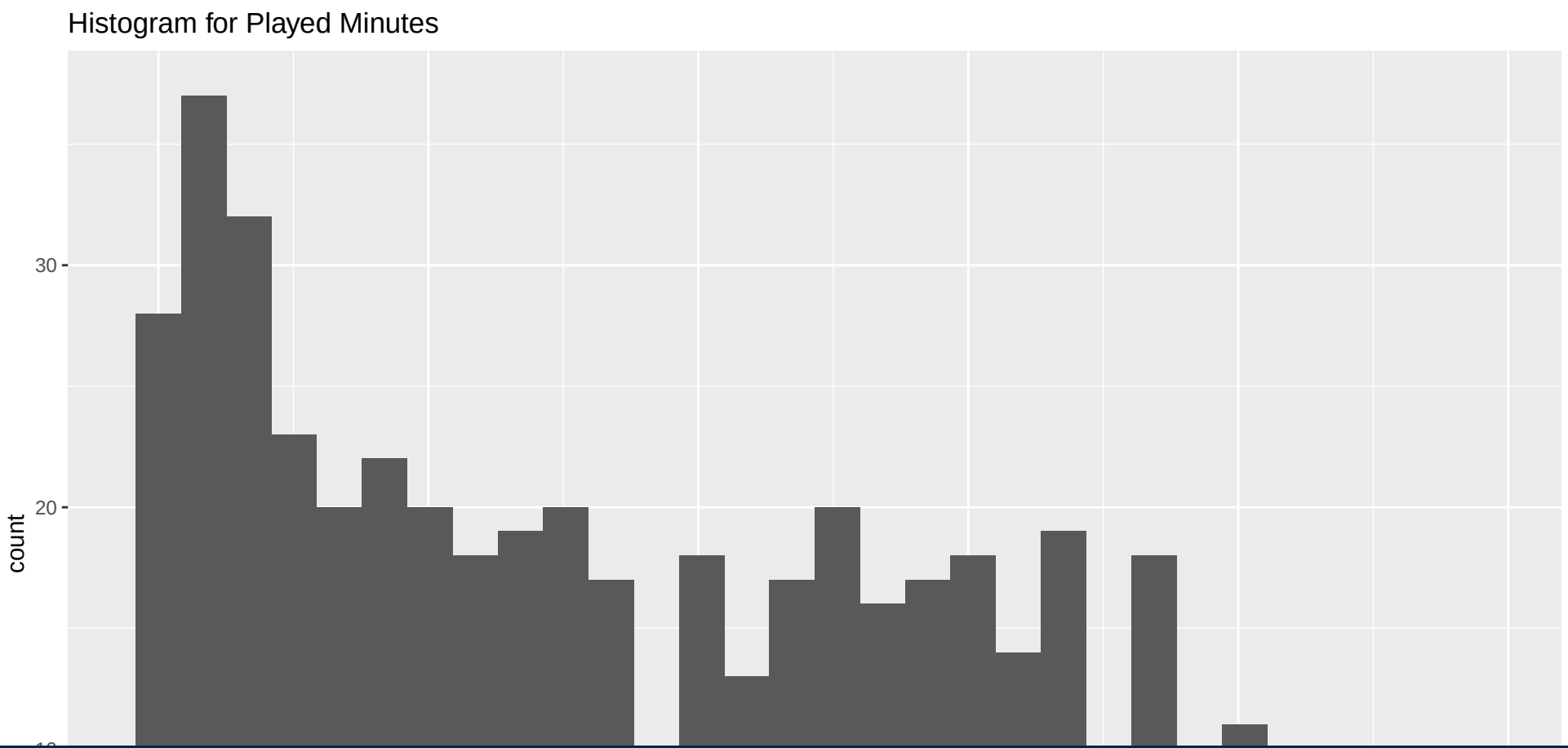
```
##     Benson, Keith        Diogu, Ike   N'Diaye, Hamady    Williams, Sean
##         2.1176032         1.8681832         1.5567575         1.0007997
## Whiteside, Hassan    Nelson, Jameer     Bledsoe, Eric      Lopez, Brook
##         0.9188331         0.8786333         0.8486712         0.8390129
##     Orton, Daniel   Macklin, Vernon
##         0.8211618         0.8188564
```

Plus/minus for NBA

```
summary(nba12p$SimpleMin)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##     0.0   292.1   812.5   892.0  1416.6  2430.9
```

```
ggplot(data=nba12p, aes(SimpleMin))+geom_histogram()+
  ggtitle("Histogram for Played Minutes")
```



Plus/minus for NBA

Getting players with more than 250m of play

```
more250<-nba12p$PlayerID[nba12p$SimpleMin>250]
length(more250)
```

```
## [1] 372
```

```
more250[1:10]
```

```
##  [1] "17"  "51"  "109" "132" "144" "211" "459" "484" "514" "550"
```

```
# Subset the dataframe by more250
colnames(nba5) <- gsub("X", "", colnames(nba5))
nba6<-nba5[,colnames(nba5) %in% more250]
nba6<-data.frame(PM=nba5$PM, nba6)
# Make the model
model2<-lm(PM~., data=nba6)
# get the coefficients
coeff2<-sort(coefficients(model2), decreasing=T)
coeff2[1:10]
```

```
##      X158      X113      X807       X70      X144      X904      X821      X791
## 0.8779187 0.7791911 0.7605357 0.6760659 0.6705972 0.6580038 0.6435656 0.6088434
##      X938        X5
## 0.6001202 0.5689375
```

Plus/minus for NBA

```
name1<-names(coeff2)
name1<-gsub("X", "", name1)
name1<-as.data.frame(name1)
name2<-left_join(name1, nba12p[,1:2], by=c("name1"="PlayerID"))
names(coeff2)<-name2$PlayerTrueName
```

```
coeff2[1:10]
```

```
## Nelson, Jameer   Duhon, Chris Hudson, Lester    Paul, Chris   Battie, Tony
##      0.8779187      0.7791911      0.7605357      0.6760659      0.6705972
##  Bledsoe, Eric Curry, Stephen   Lopez, Robin     Wall, John Nowitzki, Dirk
##      0.6580038      0.6435656      0.6088434      0.6001202      0.5689375
```

Plus/minus for NBA

Where is Durant ?

```
name2[name2$PlayerTrueName=="Durant, Kevin",]
```

```
##     name1 PlayerTrueName
## NA   <NA>           <NA>
## 225   714  Durant, Kevin
```

Where is Lebron ?

```
name2[name2$PlayerTrueName=="James, LeBron",]
```

```
##    name1 PlayerTrueName
## 15   260  James, LeBron
## NA  <NA>           <NA>
```

Plus/minus for NBA

```
str(nba12)
```

```
## 'data.frame':    25300 obs. of  36 variables:
##  $ GameID          : Factor w/ 883 levels "20111225BOSNYK",..: 1 1 1 1 1 1 1 1 1
##  $ Date            : chr  "20111225" "20111225" "20111225" "20111225" ...
##  $ AwayT           : chr  "BOS" "BOS" "BOS" "BOS" ...
##  $ HomeT           : chr  "NYK" "NYK" "NYK" "NYK" ...
##  $ StartTime       : chr  "0:48:00" "0:42:59" "0:42:17" "0:41:42" ...
##  $ EndTime         : chr  "0:42:59" "0:42:17" "0:41:42" "0:40:35" ...
##  $ ElapsedTime     : chr  "0:05:01" "0:00:42" "0:00:35" "0:01:07" ...
##  $ ElapsedSecs     : int  301 42 35 67 58 35 146 30 6 165 ...
##  $ HomePlayer1ID   : int  843 843 843 990 990 990 990 990 990 843 ...
##  $ HomePlayer2ID   : int  915 915 915 915 752 752 752 25 25 990 ...
##  $ HomePlayer3ID   : int  25 25 25 25 25 25 25 315 315 315 ...
##  $ HomePlayer4ID   : int  537 537 537 315 315 315 315 597 597 537 ...
##  $ HomePlayer5ID   : int  118 118 118 118 118 988 988 988 988 597 ...
##  $ AwayPlayer1ID   : int  553 553 553 553 553 553 876 876 553 553 ...
##  $ AwayPlayer2ID   : int  166 166 166 166 419 419 419 419 876 876 ...
##  $ AwayPlayer3ID   : int  265 14 14 14 14 14 14 14 14 14 ...
##  $ AwayPlayer4ID   : int  285 83 181 181 181 181 181 181 181 181 ...
##  $ AwayPlayer5ID   : int  143 285 83 83 83 83 285 285 285 285 ...
##  $ HomePlayer1Name : chr  "Douglas" "Douglas" "Douglas" "Shumpert" ...
##  $ HomePlayer2Name : chr  "Fields" "Fields" "Fields" "Fields" ...
##  $ HomePlayer3Name : chr  "Anthony" "Anthony" "Anthony" "Anthony" ...
##  $ HomePlayer4Name : chr  "Stoudemire" "Stoudemire" "Stoudemire" "Jeffries" ...
```

Plus/minus for NBA

PM for possessions #getting the PM for possessions

```
nba12$PM_Home<-nba12$PointsScoredHome/nba12$PossessionsHome
nba12$PM_Away<-nba12$PointsScoredAway/nba12$PossessionsAway
nba12$Margin<-nba12$PM_Home-nba12$PM_Away
summary(nba12$PM_Home)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
##    0.00    0.50    1.00     Inf    1.62     Inf    2085
```

```
summary(nba12$PM_Away)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
##  0.0000  0.4444  1.0000     Inf  1.5000     Inf    2120
```

```
summary(nba12$Margin)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
##    -Inf  -0.667   0.000     NaN   0.692     Inf    3576
```

We have choice, remove cases with not number values, or impute with the mean

We will remove

```
nba6$PM<-nba12$Margin
nba6$Secs<-nba12$ElapsedSecs/60
nba7<-nba6[is.finite(nba6$PM),]
dim(nba7)
```

Plus/minus for NBA

Where is Durant ?

```
name2[name2$PlayerTrueName=="Durant, Kevin",]
```

```
##      name1 PlayerTrueName
## 102    714  Durant, Kevin
## NA    <NA>           <NA>
## NA.1  <NA>           <NA>
```

Where is LeBron ?

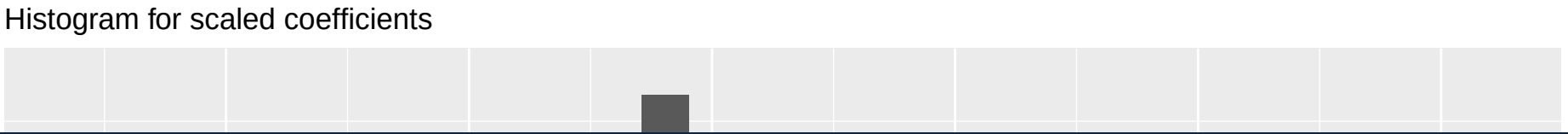
```
name2[name2$PlayerTrueName=="James, LeBron",]
```

```
##      name1 PlayerTrueName
## 20     260  James, LeBron
## NA    <NA>           <NA>
## NA.1  <NA>           <NA>
```

Scaling the rating

```
coeff3<-coeff3-mean(coeff3)
```

```
ggplot()+aes(coeff3)+geom_histogram()+
  ggtitle("Histogram for scaled coefficients")
```



Plus/minus for NBA

```
load("./data/nba12.rda")
```

```
nba7$Secs<-nba7$Secs/60
model4<-lm(PM~.,data=nba7, weights = Secs)
coeff4<-sort(coefficients(model4), decreasing=T)
# basic cleaning
name1<-names(coeff4)
name1<-gsub("X", "", name1)
name1<-as.data.frame(name1)
name2<-left_join(name1, nba12p[,1:2], by=c("name1"="PlayerID"))
names(coeff4)<-name2$PlayerTrueName
coeff[1:10]
```

```
##     Benson, Keith         Diogu, Ike   N'Diaye, Hamady    Williams, Sean
##         2.1176032          1.8681832          1.5567575         1.0007997
## Whiteside, Hassan    Nelson, Jameer     Bledsoe, Eric      Lopez, Brook
##         0.9188331          0.8786333          0.8486712         0.8390129
##     Orton, Daniel   Macklin, Vernon
##         0.8211618          0.8188564
```